

06 BANK DATABASE – TRIGGERS AND STORED PROCEDURES

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1. Procedure — Branch Accounts

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: 06-TRIGGERS-STORED-PROC...

SCHEMAS

Filter objects

bookflow_db

sys

83 SELECT *

84 FROM Account

85 WHERE Branch = p_Branch;

86 END //

87

88 DELIMITER ;

89

90 CALL GetBranchAccounts('Hyderabad');

Result Grid

Filter Rows: Export: Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad

Administration Schemas

Information

No object selected

Result 1 x

Output

Action Output

#	Time	Action	Message
8	12:17:03	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	5 row(s) affected Records: 5
9	12:17:06	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5
10	12:17:08	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, '...	4 row(s) affected Records: 4
11	12:17:42	CALL GetBranchAccounts('Hyderabad')	Error Code: 1305. PROCEDU
12	12:17:54	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM ...	0 row(s) affected
13	12:17:56	CALL GetBranchAccounts('Hyderabad')	3 row(s) returned

Object Info Session

2. Trigger — Prevent Negative Balance

The screenshot shows the MySQL Workbench interface with a query editor window titled "06-TRIGGERS-STORED-PROC...". The query contains the following SQL code:

```
90 • CALL GetBranchAccounts('Hyderabad');
91
92 DELIMITER //
93
94 • CREATE TRIGGER CheckBalance
95 BEFORE UPDATE ON Account
96 FOR EACH ROW
97 BEGIN
98     IF NEW.Balance < 0 THEN
99         SIGNAL SQLSTATE '45000'
100         SET MESSAGE_TEXT =
101             'Transaction failed: Insufficient balance';
102     END IF;
103 END //
104
105 DELIMITER ;
106
107 • UPDATE Account
108 SET Balance = Balance - 60000
109 WHERE Account_No = 10001;
```

The Output window shows the execution results:

#	Time	Action	Message
10	12:17:08	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, '...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
11	12:17:42	CALL GetBranchAccounts(Hyderabad)	Error Code: 1305. PROCEDURE bankdb.GetBranchAccounts dc
12	12:17:54	CREATE PROCEDURE GetBranchAccounts(IN p_Branch VARCHAR(50)) BEGIN SELECT * FROM ...	0 row(s) affected
13	12:17:56	CALL GetBranchAccounts(Hyderabad)	3 row(s) returned
14	12:18:55	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW Bal...	0 row(s) affected
15	12:18:58	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance

3. Trigger — Transaction Audit

The screenshot shows the MySQL Workbench interface with a query editor window titled "06-TRIGGERS-STORED-PROC...". The query contains the following SQL code:

```
142 DELIMITER ;
143
144 • INSERT INTO Bank_Transaction
145 (Account_No, Transaction_Type, Amount)
146 VALUES
147 (10001, 'DEPOSIT', 2500);
148
149 • SELECT * FROM Transaction_Audit;
```

The Result Grid shows the following data:

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
1	1	10001	DEPOSIT	2500	2018-12-21 12:21:38

The Output window shows the execution results:

#	Time	Action	Message
13	12:17:56	CALL GetBranchAccounts(Hyderabad)	3 row(s) returned
14	12:18:55	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW Bal...	0 row(s) affected
15	12:18:58	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Trans...
16	12:20:05	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID...	0 row(s) affected
17	12:21:26	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN IN...	0 row(s) affected
18	12:21:38	SELECT * FROM Transaction_Audit LIMIT 0, 1000	0 row(s) returned

4. Trigger — Automatic Deposit Balance Update

```
06-TRIGGERS-STORED-PROC... x
Limit to 1000 rows
174 • INSERT INTO Bank_Transaction
175   (Account_No, Transaction_Type, Amount)
176   VALUES
177   (10001, 'DEPOSIT', 5000);
178
179 • SELECT *
180   FROM Account
181   WHERE Account_No = 10001;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 3 x

Output

Action Output

#	Time	Action	Message
✓ 16	12:20:05	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID...	0 row(s) affected
✓ 17	12:21:26	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN IN...	0 row(s) affected
✓ 18	12:21:38	SELECT * FROM Transaction_Audit LIMIT 0, 1000	0 row(s) returned
✓ 19	12:22:58	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH RO...	0 row(s) affected
✓ 20	12:23:09	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 50...	1 row(s) affected
✓ 21	12:23:11	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

5. Procedure — Money Transfer

```
06-TRIGGERS-STORED-PROC... x
Limit to 1000 rows
216
217   DELIMITER ;
218
219 • CALL TransferMoney(10001, 10002, 5000);
220
221 • SELECT *
222   FROM Account
223   WHERE Account_No IN (10001, 10002);
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	80000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 4 x

Output

Action Output

#	Time	Action	Message
✓ 19	12:22:58	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH RO...	0 row(s) affected
✓ 20	12:23:09	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 50...	1 row(s) affected
✓ 21	12:23:11	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 22	12:24:12	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected
✓ 23	12:24:17	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
✓ 24	12:24:20	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned